

ELAKKIYA SELVARAJAN

React Native Engineer | Frontend Developer | Mobile (Android & iOS)

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PROFESSIONAL SUMMARY

Results-driven React Native Engineer with **2+ years of production experience** shipping cross-platform mobile apps to the **Google Play Store and Apple App Store** at an IIT Madras ICSR incubated startup. Delivered measurable impact: **30% fewer QA defects, 25% faster feature delivery, and 40% improvement in render performance**. Deep hands-on expertise in TypeScript, React Native CLI, native module bridging (Swift/Kotlin), offline-first architecture, Redux Toolkit, and CI/CD pipelines. Experienced in the full mobile development lifecycle from architecture design through App Store release.

TECHNICAL SKILLS

Mobile Development: React Native CLI, Expo, TypeScript, JavaScript, React Navigation, AsyncStorage, Native Module Bridging

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design

State Management: Redux, Redux Toolkit, Context API

Backend & Databases: NestJS, REST APIs, Axios, SQLite, PostgreSQL

Native & Platform: Kotlin, Swift, Objective-C Bridge, WiFi Direct, TCP Sockets, Offline-First Architecture

Testing & Tools: Jest, Git, GitHub, Postman, Android Studio, Xcode, VS Code, Jira, Antigravity

Practices: Agile/Scrum, CI/CD, Performance Optimization, Cross-Platform Development, Code Review

EXPERIENCE

Frontend Developer & React Native Engineer | *Plenome Technologies Pvt. Ltd.*

Oct 2023 – Present

IIT Madras ICSR Incubated Startup • Chennai, India

Aayush – Health & Wellness App (Live on Google Play Store & Apple App Store)

- Architected and shipped **10+ production screens** for Android and iOS using React Native CLI and TypeScript, reducing QA-reported defects by **30%** through reusable component design patterns.
- Engineered **20+ reusable TypeScript UI components**, cutting new feature development time by **25%** while maintaining a consistent cross-platform design system.
- Integrated **6+ REST APIs** via Axios with centralized error handling, retry logic, and loading state management, improving data reliability and perceived performance.
- Implemented **Apple Watch integration** via native module bridging (Swift + Objective-C bridge), enabling real-time health metric sync between iPhone and Watch.
- Applied React.memo, useMemo, and useCallback across critical render paths — **reduced unnecessary re-renders by 40%** and improved frame rates on low-end Android devices.
- Led **React Native version migration from v0.73 to v0.81**, resolving 15+ breaking dependency conflicts to maintain full Android and iOS compatibility.
- Built secure custom form validation across **8+ modules** using TypeScript, significantly reducing invalid submissions and improving input reliability.
- Debugged platform-specific runtime crashes, native bridge errors, and rendering issues across both Android and iOS build environments.
- Delivered features consistently on schedule within a **5-member Agile team** using Git, GitHub, and Jira with bi-weekly sprints.

Offline eRupee – Offline Peer-to-Peer Digital Payment SDK

- Designed a proof-of-concept **offline P2P digital payment system** enabling eRupee transactions without internet connectivity using React Native and TypeScript.
- Built an **Android native module in Kotlin** leveraging WiFi Direct for peer device discovery and TCP socket-based data transfer, integrated into React Native via NativeModules bridge.

- Implemented **SQLite offline transaction queue** with automatic cloud sync on network restoration, ensuring zero data loss across sessions.
- Architected a **modular, distributable React Native SDK** with a clean public API for seamless integration into external apps as a local package dependency.
- Designed **cryptographic transaction validation** workflows to prevent duplicate and invalid offline payments.
- Configured **Android Gradle multi-module build system** with React Native autolinking for SDK packaging and distribution.

React Native Developer – Intern | *Plenome Technologies Pvt. Ltd.*

Feb 2023 – Sep 2023

- Developed responsive UI components and screens using React Native, HTML5, CSS3, and JavaScript, meeting design specifications and cross-browser compatibility requirements.
- Validated REST APIs using Postman, ensuring smooth frontend-backend integration during feature development cycles.
- Collaborated with senior developers to debug UI issues, implement design feedback, and deliver production-ready frontend features.

Web Developer – Intern | *Compunet Connections*

Jun 2021 – Nov 2021

- Built and maintained responsive web pages using HTML5, CSS3, and JavaScript, adhering to design specifications and cross-browser compatibility standards.
- Supported REST API integration and validation using Postman in collaboration with backend developers.

PROJECTS

Finguard AI – Full Stack AI Financial Tracker

Mar 2026 – Present

Next.js • Tailwind CSS • NestJS • PostgreSQL • HuggingFace AI • TypeScript

- Building a full-stack AI-powered personal finance platform with intelligent expense analysis and financial insights using **HuggingFace AI models**.
- Designing responsive dashboards with **Next.js and Tailwind CSS**; architecting RESTful backend services with NestJS and PostgreSQL for scalable data management.
- Implementing personalized AI recommendations for spending pattern analysis, enabling data-driven financial decisions.

EDUCATION

M.Sc. (Integrated) – Computer Science

Aug 2018 – Jul 2023

College of Engineering, Guindy (CEG), Anna University, Chennai | CGPA: 7.67 / 10.0 | 5-Year Integrated Master's Program

CERTIFICATIONS

- **Applied Database System Engineering** – IITM Pravartak | May 2025 – Jul 2025 | Distributed databases, kernel-level processing, query optimization
- **React Native – The Practical Guide** – Udemy | March 2026 | React Native, Cross-Platform Mobile Development